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# CV

My Professional CV



## Contact Information

|                    |                              |
|--------------------|------------------------------|
| Name               | Masoud Rezvani               |
| Professional Title | ML Engineer & Data Scientist |
| Email              | s.masoudrezvani@gmail.com    |
| Location           | Amsterdam,                   |

## Professional Summary

ML Engineer and Data Scientist with 5+ years of production experience spanning fraud detection, RAG systems, and LLM engineering. Currently completing an MSc in Data Science at UvA. Published in Expert Systems with Applications (ESWA, 2025). Sweet spot: read a paper, turn it into production-ready code, and ship it.

## Experience

2026 - PRESENT



### AI Automation Intern

Talk360

*International calling service (Amsterdam).*

- Built an HR policy assistant using hybrid retrieval (ChromaDB semantic + BM25 keyword) reranked by a Cross-Encoder, achieving 98% faithfulness on a 22-question golden dataset; added inline source citations via system prompt design and an LLM-based prompt-injection firewall.
- Built a semantic caching layer to reduce repeat-query API costs; implemented a custom LLM-as-judge pipeline (GPT-4o) to track regression before deploying new code.
- Built autonomous LangChain + Gemini agents to interpret execution logs and diagnose system failures.
- Engineered n8n ETL pipelines ingesting Mixpanel usage metrics and routing anomaly alerts to Slack.

2024 - 2025



### Data Scientist

Baly.iq (Rocket Internet)

*The #1 super app in Iraq.*

- Built and tuned XGBoost and logistic regression models for fraud detection, reaching 95% accuracy and contributing to \$500K in estimated annual savings.
- Built reproducible Airflow and Python ETL pipelines to process unstructured fraud data, cutting manual work by 30% and making updates repeatable across environments.
- Led a team of analysts to improve fraud ring detection by 10% using GNNs; built a graph visualization tool that surfaced fraudulent connections, saving ~\$1,000 daily.
- Deployed containerized fraud-detection services with Docker and Ansible across multi-server environments; tuned MySQL and

PostgreSQL query performance with Percona PMM, improving response time by 20%.

- Maintained AWS EC2 server environments: configured monitoring (Grafana, Netdata), managed pipeline scheduling (Airflow, cronjobs), and resolved network issues (iptables, DNS).

2020 - 2024



### Data Analyst

Snapp! (Rocket Internet)

*The #1 ride-hailing super app in Iran.*

- Investigated raw logs, relational tables, and metadata to design SQL and Python detection rules, identifying anomalies 40% faster than manual review.
- Applied CNNs and GNNs to detect complex fraud patterns and fraud rings, improving detection efficiency by 10%.
- Built tagging logic for fraud categories and structured irregular transaction flows into ML-ready datasets; collaborated with downstream modeling teams on feature curation.
- Led collaborative fraud case reviews with operations and legal teams to surface business logic loopholes across driver, passenger, and payment channels.
- Self-taught and implemented Airflow and Docker pipelines to automate fraud-data workflows, later forming the production foundation at Baly.iq.

## Education

2025 - 2026



### Master of Science

University of Amsterdam (UvA)

*Data Science*

- Thesis: Evaluating the Muon Optimizer — From Symbolic Regression Geometry to High-Dimensional LLM Adaptation

2017 - 2019



### MBA

University of Tehran

*Business Administration*

2012 - 2016



### BSc

Amirkabir University of Technology (Tehran Polytechnic)

*Industrial Engineering*

## Publications

2025

### [WDAE-GAN: A Hybrid Dual Autoencoder and Generative Adversarial Framework with Wavelet Denoising for Credit Card Fraud Detection](#)

*Expert Systems with Applications (ESWA)*

Developed a Wasserstein GAN and dual autoencoder framework with wavelet denoising to solve severe class imbalance in financial fraud datasets. Achieved near-perfect detection accuracy while maintaining extremely low false-positive rates.

## Skills

**ML Frameworks:** PyTorch, HuggingFace, scikit-learn, XGBoost

**MLOps & LLMOps:** MLflow, FastAPI, Streamlit, Docker, Airflow, LangChain, Git

**Cloud & Databases:** AWS, ChromaDB, PostgreSQL, MySQL

## Languages

**Persian :** Native speaker

**English :** Fluent

**Dutch :** Beginner

## Interests

💡 **Research Interests:** Large Language Models, Deep Learning, Generative AI, Graph Neural Networks, MLOps

## Certificates

**AI Engineer Core Track (LLM Engineering, RAG, QLoRA, Agents) - Udemy (2026)**

**Machine Learning System Design - ByteByteGo (2026)**

**Stanford CME295 — Transformers and Large Language Models - Stanford University (2026)**

**Stanford CS336 — Language Modeling from Scratch - Stanford University (2025)**

**Associate DevOps Engineer - Arvancloud (2025)**

**Graph Search, Shortest Paths, and Data Structures (Algorithms Specialization, Part 2) - Coursera (2024)**

**Applied Statistical Modeling for Data Analysis in R - Udemy (2023)**

## Projects

### Vision-Language Model for DocVQA

- Fused visual embeddings (ResNet + CLIP) with OCR text outputs fed into a transformer LLM.
- Evaluated on standard DocVQA benchmarks; trained end-to-end in PyTorch.

### SemArt: Texture Bias in CNNs on Historical Art

- Benchmarked ResNet-50 and CLIP using InfoNCE contrastive learning.
- Quantified texture bias via UMAP visualization and inter-class distance metrics.

### Interactive Model Interpretation Dashboard

- Backend built with FastAPI; frontend exposes real-time SHAP dependency plots.
- Fully containerized with Docker for reproducible deployment.

### Algorithmic Trading System

- Feature engineering over price-action patterns (support/resistance, volume profiles, candlestick structures) with Polars

- All experiments tracked with Weights & Biases for systematic ablations across feature sets and models.

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#### **LLM Arabic Fraud Detection**

- Engineered data-cleaning and tokenization strategies suited for Arabic morphology.
- Built strict evaluation pipelines for reproducible, production-ready results.

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#### **CARE-GNN Reconstruction**

- Full reconstruction of heterogeneous graph construction, label-aware similarity estimation, and RL-based neighbour selector.
- Surfaced underspecified implementation details around camouflage-aware aggregation and the RL exploration-exploitation tradeoff.